

Main Results

Method	Category	Seeds	Mean reward	Bitrate	Rebuffering	Smoothness
BBA rule-based algorithm	Traditional	1	0.7642 +/- 0.0000	1183.3 +/- 0.0	0.0472 +/- 0.0000	0.3974 +/- 0.0000
PC predictive control algorithm	Traditional	1	0.8485 +/- 0.0000	1212.9 +/- 0.0	0.0953 +/- 0.0000	0.1375 +/- 0.0000
reinforcement learning baseline	RL baseline	5	0.7509 +/- 0.0347	1096.9 +/- 3.4	0.0834 +/- 0.0081	0.1684 +/- 0.0019
RL reinforcement learning baseline	RL baseline	5	0.8587 +/- 0.0243	1104.8 +/- 6.3	0.0792 +/- 0.0043	0.0885 +/- 0.0032
semantic reprogramming baseline	This work	5	0.8068 +/- 0.0804	1030.4 +/- 69.7	0.0765 +/- 0.0210	0.0764 +/- 0.0062
text pre-alignment enhancement	This work	5	0.8547 +/- 0.1000	1015.8 +/- 26.7	0.0627 +/- 0.0225	0.0746 +/- 0.0101
text multi-scale history enhancement	This work	5	0.8967 +/- 0.0475	1057.0 +/- 33.7	0.0610 +/- 0.0114	0.0821 +/- 0.0241